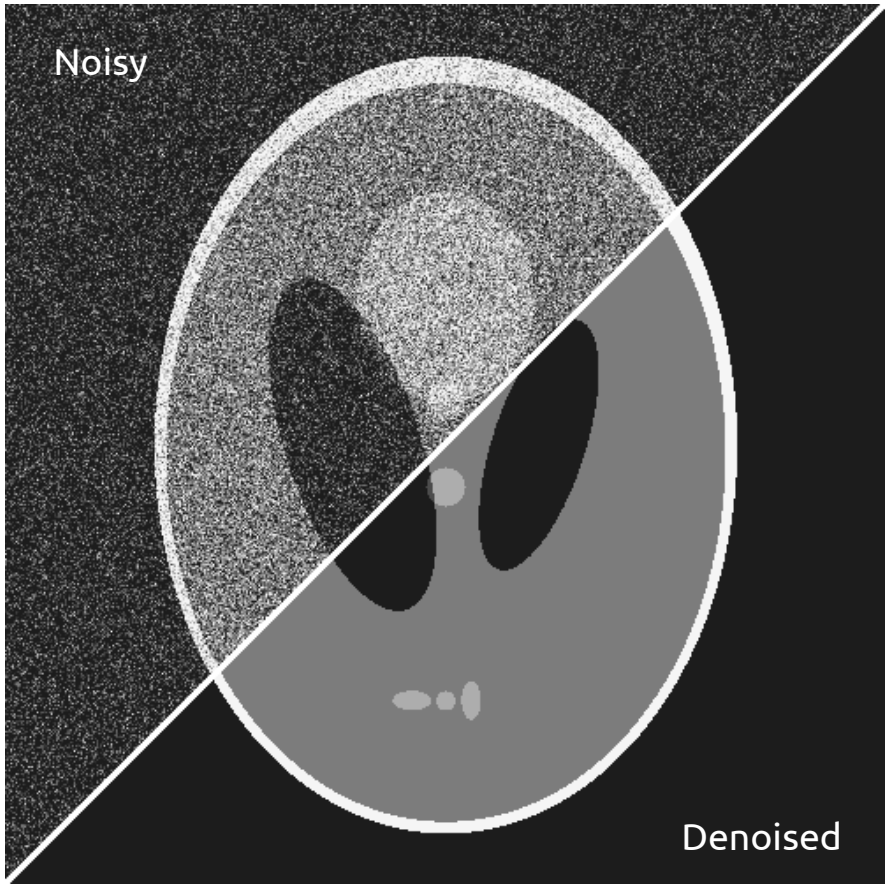
Amsterdam Imaging Hub

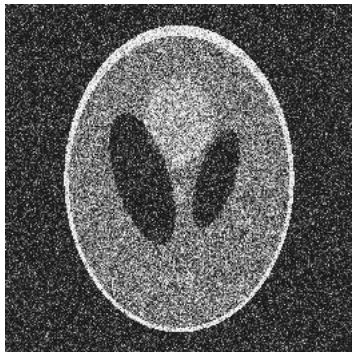
Neural networks for image denoising

Alexander Skorikov
Jiayang Shi



What is image denoising?

- **Goal:** recover the true image x from a noisy measurement y
- **Ill-posed:** noise erases information, many images x fit the same y
- **Way out:** add prior knowledge of the object, noise and measurement

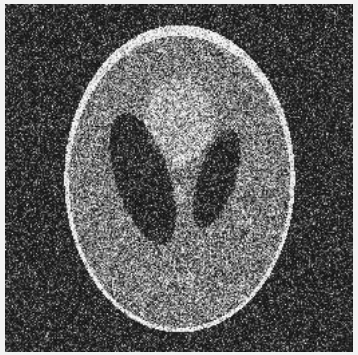


Measurement y

$y = x + n$

recover x?

Many images x fit the same y



Keeps all the noise



Smooths away detail



Plausible under prior

Types of image noise/artifacts

Random noise (improves with averaging/exposure)



Poisson

Fundamental statistics: intensity-dependent



Gaussian

Electronics, heat: same spread everywhere



Correlated

Scan lines, stripes: errors shared by neighbors

Deterministic distortions



Blur

Optics, motion: neighboring pixels mixed



Detector limits

Saturation and rounding of the reading

Real data usually mix several of these. The noise model is part of the prior.

Neural networks

In a nutshell: an effective method to **fit complex, multidimensional functions** that **generalizes well to unseen inputs**

Input x
e.g. a noisy image: one value per pixel, ~10^6 dimensions

→

Network $f_\theta(x)$
layers of simple operations with millions of parameters θ

→

Output y
e.g. the denoised image

Same idea as curve fitting

Least squares fits **y = ax + b** with 2 parameters; a network fits millions

Learned from examples

θ is tuned to minimize the error between the output and known **clean targets**

Generalizes

A well-trained network works on **new images** it never saw in training

Brief history

1943–1969

Early foundations

McCulloch & Pitts model a neuron as a simple threshold unit (1943)

Rosenblatt's perceptron learns its weights from data (1958)

1970s–2000s

AI winter

Minsky & Papert show the limits of single-layer perceptrons (1969); funding dries up

Backpropagation (1986) and LeCun's convolutional nets for digits (1989) keep the field alive

2012

AlexNet

A deep convolutional network trained on GPUs wins the ImageNet image-recognition contest

Top-5 error of 15% vs 26% for the runner-up

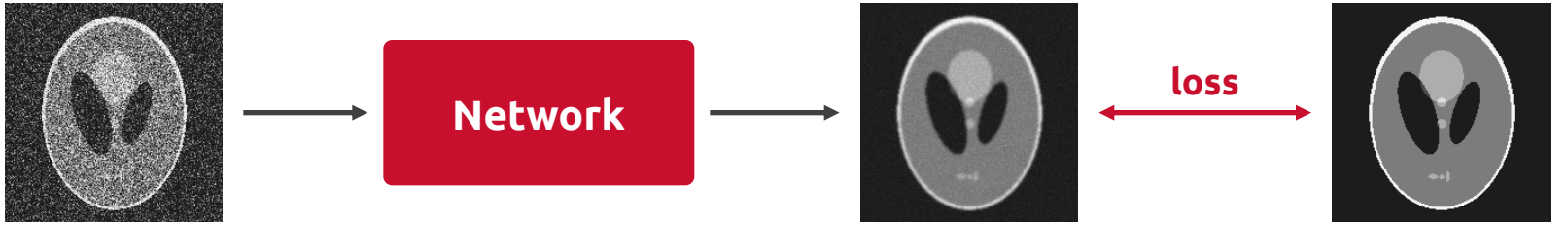
2022

GPT moment

ChatGPT reaches an estimated 100 million users in two months

Neural networks become an everyday tool

Supervised methods



Noisy y Output $\hat{x}$ Clean target x

Training: adjust the network until its output matches the clean target, over thousands of pairs

What is the training data?

Paired real data

Same sample imaged twice: short and long exposure, or low and high dose

+ Typically best performance
– Costs time; may be impossible because of sample damage or drift

Simulated data

Clean images or phantoms plus noise from a physical model

+ Unlimited data, exact ground truth
– Objects of interest are often very difficult to simulate

Transfer learning

Start from a network trained on another dataset, then fine-tune

+ Needs only a few pairs of your own
– Mismatch in noise or structures

Learned-prior methods

No pairs needed: one clean-image prior works across instruments and noise levels

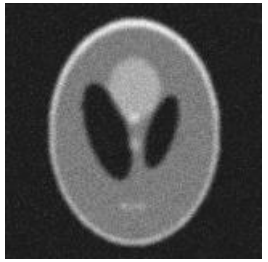


Noisy y

Prior + noise model

Prior learned from clean images

iterate



Output $\hat{x}$

Key idea: the prior pulls toward clean-looking images; the noise model keeps the result consistent with y.

Plug-and-play

Iterate between neural denoising and data fit steps

+ Reuses any trained denoiser; doesn't need many network passes
– Step size and iteration count need tuning

Diffusion posterior sampling

A diffusion model trained on clean images generates the output, guided by y at every step

+ State-of-the-art generative model; uncertainty quantification
– Slow: hundreds of network passes; can invent structure

Self-supervised methods

No clean images needed: the network learns from noisy data alone

Why it works: noise changes between images and pixels, the signal does not, so the network can only learn to predict the signal
Assumption: noise must be zero-mean and random; blur and other systematic distortions stay in the image

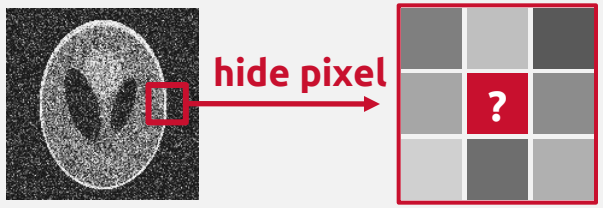
Noise2Noise



Two noisy images of the same sample: one is the input, the other the target

+ Close to supervised; pairs often free (even/odd frames)
– Noise must be independent between the two; static sample

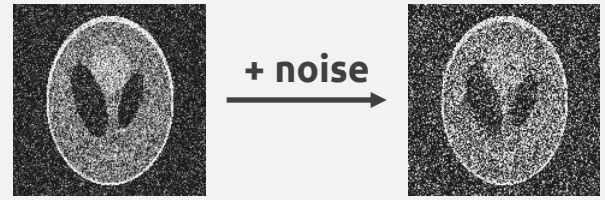
Noise2Void / Noise2Self



Hide a pixel and predict it from its neighbours, using single noisy images

+ Train directly on the images you want to denoise
– Blurrier than N2N; fails on correlated noise

Noisier2Noise / R2R

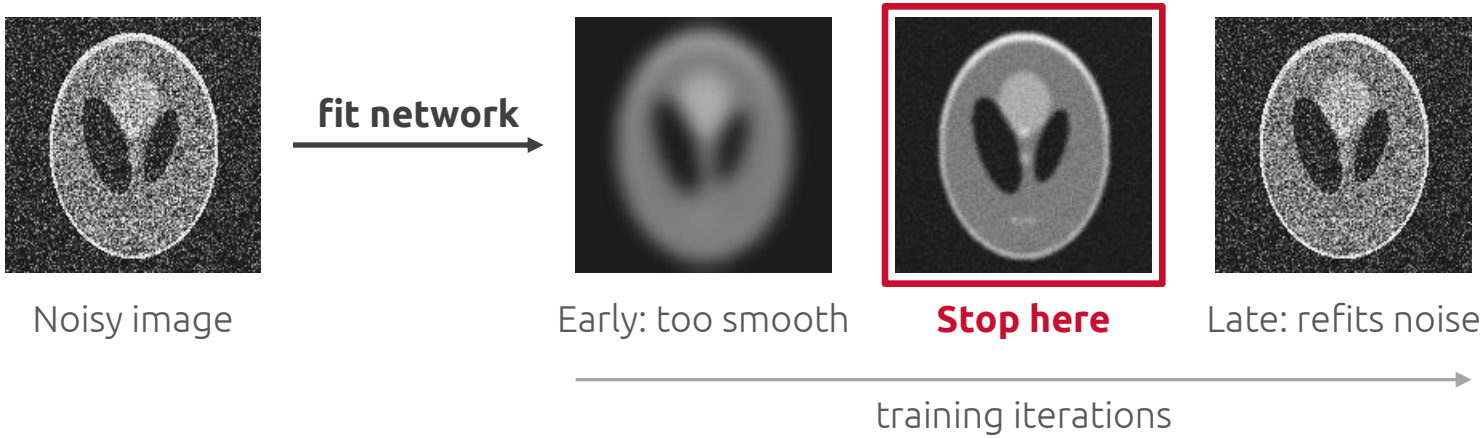


Add extra noise from a known model and learn to undo it, on single images

+ No blind spot: sharper, can model correlated noise
– Needs known noise type and level

Zero-shot methods

No training set: the network learns from the single noisy image you want to clean



Key idea: networks fit image structure before they fit noise. Stop early and you keep the structure.

Zero-shot N2N / Self2Self

Split the one image into sub-images or random masks, then train self-supervised on it alone

+ Small networks, fast to fit
– Assumes pixel-independent noise

Implicit neural representations

A small network maps pixel coordinates (x, y) to intensity; it learns smooth structure first

+ Continuous image at any resolution; can fit raw measurements, e.g. tomography projections
– Network size and frequency settings act as the prior and need tuning



What method to choose?

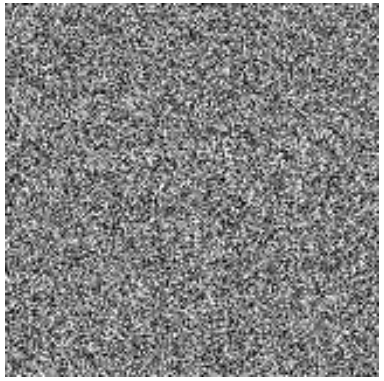
Mostly depends on the data you have. Higher rows need more data and usually give better results:

If you have…	Use	Examples and watch-outs
Noisy + clean pairs e.g. short and long exposure	→ **Supervised** slide 6	CARE, DnCNN, U-Net – Needs matched pairs; struggles on new noise
Clean images + noise model e.g. reference scans, simulations	→ **Learned prior** slide 7	Plug-and-play, diffusion posterior sampling – Slow; can invent structure
Many noisy images only no clean reference	→ **Self-supervised** slide 8	Noise2Noise, Noise2Void, Noise2Self – Assumes pixel-independent noise
A single noisy image one-off or unusual sample	→ **Zero-shot** slide 9	Deep Image Prior, INRs, Self2Self – Slow per image; needs early stopping

Can you trust the output?

With neural networks “looks good” ≠ accurate! Networks can hallucinate or destroy structure.

Estimate $\hat{x}$ | Residual $y - \hat{x}$

Good
Residual is pure noise




Over-smoothed
Edges in the residual: real structure was removed

Often hard to check, but you can try to:

1. **Inspect the residual:** it should show no edges or structure
2. **Test on known truth:** simulations, phantoms or a long-exposure scan
3. **Compare independent halves:** denoise split data separately and check agreement, e.g. Fourier ring correlation
4. **Stay in distribution:** don’t trust it on samples unlike the training data

Main take-aways

1 **Denoising needs a prior**

Noise destroys information. Every method adds assumptions (sometimes implicit) about the images and the noise.

2 **Your data decides the method**

Pairs → supervised. Clean images only → learned prior. Noisy images only → self-supervised or zero-shot.

3 **Know your noise**

Is it pixel-independent? Does it depend on the signal? Most methods assume one specific model.

4 **Don’t overlook validation**

Inspect the residual and validate on known truth. Networks can invent plausible structure.



Hands-on: four notebooks

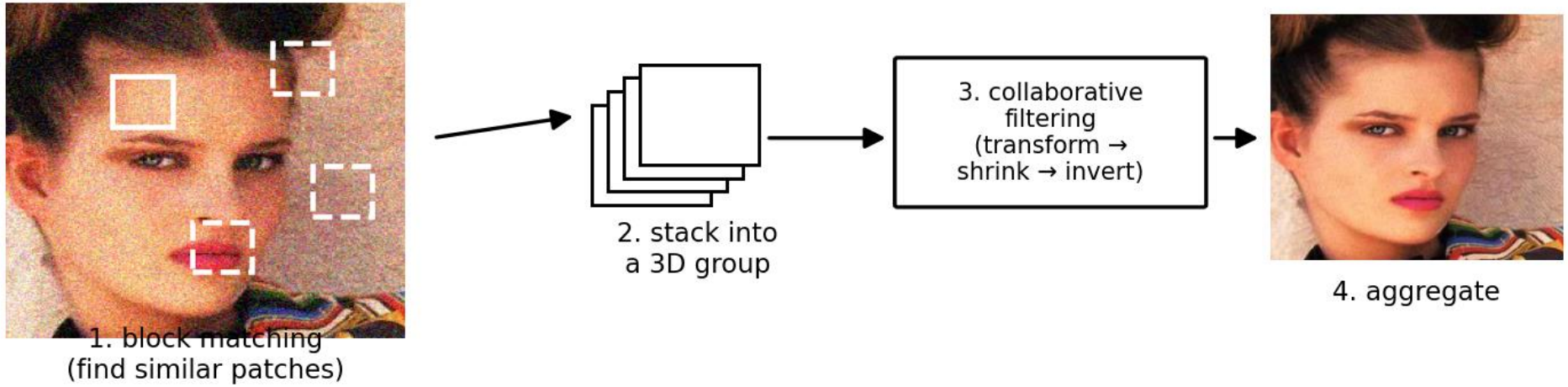
Each notebook denoises an image from a dataset of celebrity faces (CelebA) with the same Gaussian noise level ($\sigma = 25/255$), so the PSNRs compare directly

1	**BM3D**	Classical baseline
2	**Noise2Void**	Self-supervised
3	**Noise2Self**	Self-supervised
4	**DPS**	Learned prior

Notebooks: **github.com/jiayangshi/denoising-tutorial**

Exercise 1: BM3D, the classical baseline

A hand-crafted prior: natural images repeat themselves, so similar patches can be averaged



How it works

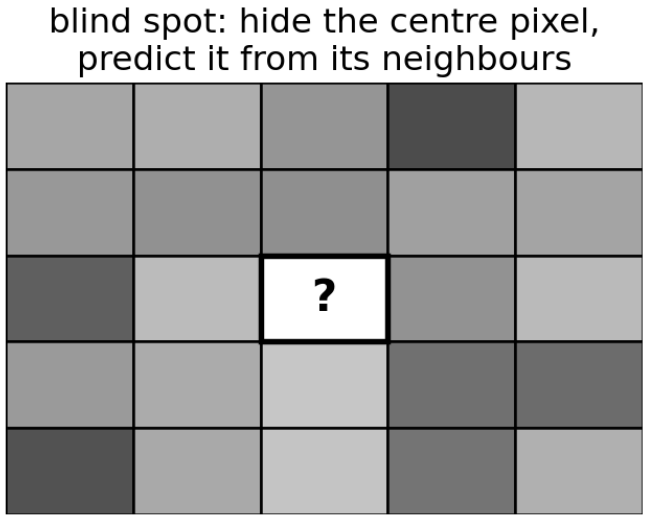
1. Block matching: find patches that look alike across the image
2. Stack each group into 3D and filter it jointly
3. Put the filtered patches back and average where they overlap

Dabov et al., "Image denoising by sparse 3-D transform-domain collaborative filtering", IEEE TIP, 2007

Notebooks: **github.com/jiayangshi/denoising-tutorial**

Exercise 2: Noise2Void

Self-supervised: the network learns from the noisy image itself, via a blind spot



How it works

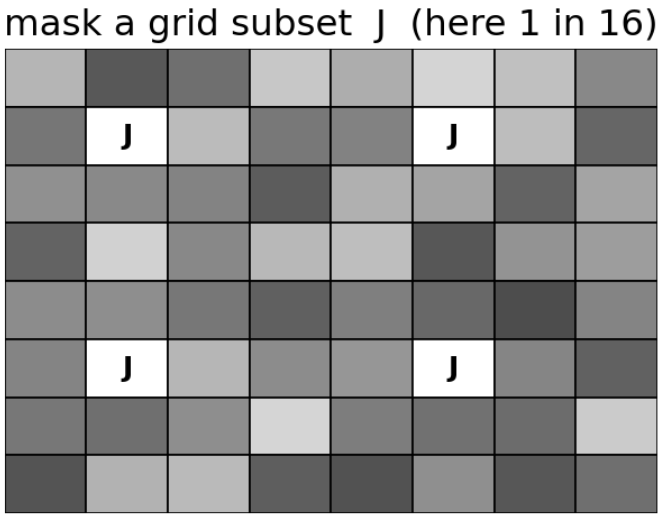
1. Hide a pixel and predict it from its neighbours only
2. Score the loss at the hidden pixel, against its noisy value
3. Signal is predictable from neighbours; independent noise is not, so it averages out

Krull, Buchholz & Jug, “Noise2Void: learning denoising from single noisy images”, CVPR, 2019

Notebooks: **github.com/jiayangshi/denoising-tutorial**

Exercise 3: Noise2Self

The same blind spot as Noise2Void, with a proof of why it denoises: J-invariance



J-invariance

prediction on J must not depend on the input values in J

minimising self-prediction error on the noisy image then also denoises the signal

How it works

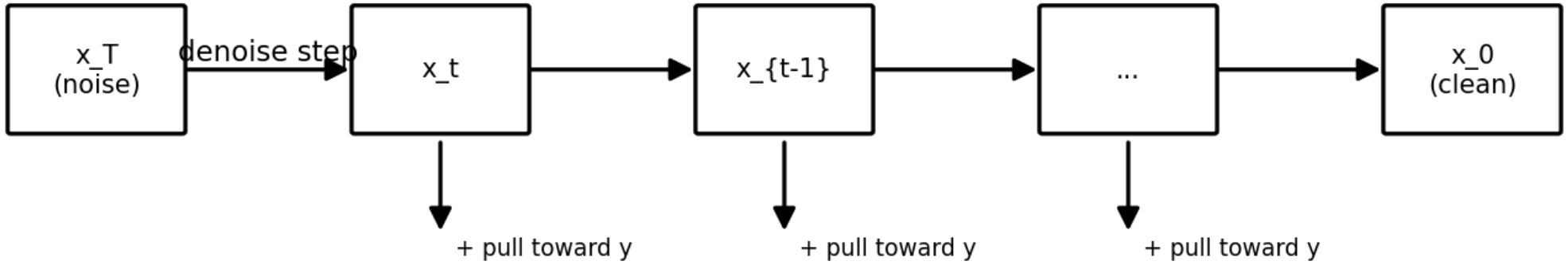
1. Mask a grid subset J of pixels (here 1 in 16)
2. The prediction on J must not depend on J's own input values
3. Then self-prediction error on the noisy image tracks error against the unseen clean image

Batson & Royer, "Noise2Self: blind denoising by self-supervision", ICML, 2019

Notebooks: **github.com/jiayangshi/denoising-tutorial**

Exercise 4: Diffusion posterior sampling

DPS is a learned prior: no training, only a diffusion model pretrained on faces



How it works

1. Start from pure noise and run the model's reverse diffusion
2. At each step, pull the estimate toward y with the gradient of $\|y - \hat{x}_0\|^2$
3. Change the operator in the data term to inpaint, upsample or deblur

Chung et al., "Diffusion posterior sampling for general noisy inverse problems", ICLR, 2023. Prior: google/ddpm-celebahq-256

Notebooks: **github.com/jiayangshi/denoising-tutorial**

DPS: prior generates, data steers

Snapshots of the reverse process, from pure noise (t = 996) to the final estimate x_0

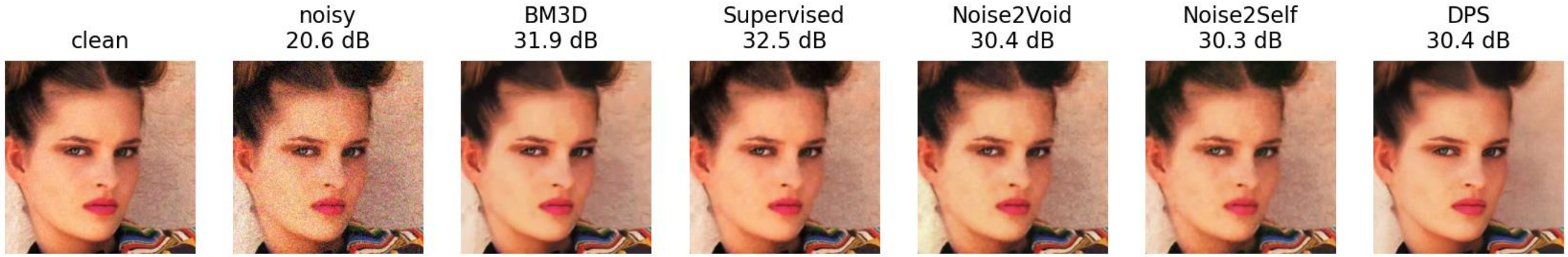


What you see: Early steps produce a generic face from the prior; later steps lock onto the measured image y.

Notebooks: **github.com/jiayangshi/denoising-tutorial**

Same face, same noise - all methods

PSNR on the same face, σ = 25/255; the noisy input scores 20.6 dB



Supervised · 32.5 dB

Best result, but needs clean pairs that most real data lack

BM3D · 31.9 dB

Close behind with no training, but needs the noise level σ

N2V, N2S, DPS · 30.3–30.4 dB

Need neither clean data nor σ; about 2 dB behind supervised

Notebooks: **github.com/jiayangshi/denoising-tutorial**

Try it on your own data

Notebooks 1–3 end with a “Try it on your own image” section

Bring an image from your own work:

1. **Upload it:** colour or single-channel (e.g. a 16-bit microscopy TIFF), any value range
2. **It is prepared for you:** normalised to [0, 1] and optionally downscaled
3. **Denoise it:** BM3D estimates σ (or set it by hand); Noise2Void and Noise2Self retrain on your image
4. **Check it:** inspect the residual and question the noise assumptions (slides 11–12)

Why not DPS?

DPS needs a diffusion prior trained on your image domain. Its face prior would turn your sample into a face (but could be fun to try 🙂).

Notebooks: **github.com/jiayangshi/denoising-tutorial**